Software Testing -Potential Exam Questions

Fortgeschrittene Themen des Softwaretests (Technische Universität München)



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QUESTIONS TESTING

Intro

1. What's the general purpose of software testing?
2. What is a test case? And what makes a good test case?
3. What is failure/error/defect? Which of them is directly addressed by testing?
4. When do you stop testing?

Requirements Testing

1. What is the purpose of requirements based testing?
2. Describe the steps at testing using cause-effect graphs. Is this testing method a good one?
3. What is an equivalence class? What is it used for?
4. Would you do requirements based testing?
5. What's the relationship between partition-based testing and requirements-based testing?
6. Describe the relationship between the following coverage criteria: Condition, Multiple Condition, Condition/Decision, MC/DC
7. How does CF based testing partition the input domain?
8. Does it make sense to do CF based testing alone? If no, why not?
9. What is the fault model behind DF based testing?
10. What is a def-use graph?
11. You need to decide between CF and DF based testing. What are the pros and cons of each of them?
12. What's the danger when using coverage criteria as a stopping criteria?

Combinatorial Testing

1. What is the motivation and failure model behind combinatorial testing?
2. How many test cases do you need to test a system with 4 parameters and 3 values (not using combinatorial testing)?
3. What are reasons to not use combinatorial testing?
4. How would you solve the problem if latin squares suggest not possible combinations for your test cases?

Random/Statistical Testing

1. How are partition testing and random testing related?
2. What's the difference between random and statistical testing?
3. What is the testing method called in which a subdomain consists of just one single element?
4. Describe the most important assumptions of the Weyuker/Jeng model. Are they justified?
5. What is the comparison between partition and random testing based on?
6. Under which circumstances is partition testing at least as good as random testing (Weyuker/Jeng)?
7. What is, in general, the effect of refining existing partitions into equally sized subdomains with same n from each of them?

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8. What does Gutjahr criticize in Weyuker/Jeng model? (Compare the models of Gutjahr and Weyuker/Jeng)
9. Describe the most important assumptions of the Gutjahr model. Are they justified?
10. What can we say about partition testing in general in comparison to random testing?
11. Are there reasons to just do random testing? In which situations would that make sense?
12. What is an operational profile?
13. Is it good to spend less test cases for functions of the system that are almost never used?
14. How are Markov Chains used to test software?

Model Based Testing

1. Which kinds of abstraction can be used for modeling?
2. What is a Mealy Machine? What are they used for?
3. What is a “good” model of the SUT and environment?
4. Describe the four scenarios on how to generate tests and code using models of the system and environment. Also describe which scenario to use if you work in a project with many suppliers and changing requirements.
5. What exactly is the advantage of having “redundancy” w.r.t. to test and code generation?
6. How do you generate test cases in MBT?
7. Is there any evidence that MBT is superior to random testing? If not, are there still good use cases for it?

OO Testing

1. What is defined as the “state of a program”?
2. If a class defines a state machine shouldn’t we simply try to cover all transitions/states in that state machine? Wouldn’t that find all failures?
3. What are Harel statecharts used for?
4. Describe three possible errors in statecharts.
5. What are Naughty Children and Worm Holes (in the context of testing of course :))?
6. What does “flattening of classes” actually mean?

From Failures to Tests and Faults

1. How do you call the process of correcting the found faults?
2. What’s the main difference between state-based and event-based test extraction?
3. Which information does a stack frame contain?
4. Describe the two variants to access stack frames to localize a fault. Which of them would you prefer and why?
5. What could be a good way to localize faults in a GUI?
6. Does Delta Debugging minimize test cases or rather localize a fault?
7. What is a global minimum w.r.t. Delta Debugging?
8. Explain the difference between Code Churn, Code Metric and Hit Spectra. Describe which of them is the worst/best and why?
9. Would you spend more test cases on a method if it has a cyclomatic complexity of 20 instead of 3?

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10. Is code reading proved to be better than testing the code?
11. Does it make sense to combine code-reading and testing?

Fault Injection and Mutation Testing

1. What is the purpose of injecting faults into code and running an existing test suite?
2. When is a test suite “good”?
3. Why is it impossible to kill equivalent mutants? Give an example of an equivalent mutant.
4. Under which assumptions does mutation testing work? Are they justified?
5. Is your test suite better if it is able to kill mutants of a higher order?
6. Explain the difference between weak and strong mutation testing.
7. Does Fault Injection increase the quality of a test suite?
8. Assume your test suite has 100% statement coverage on a given program. Does it still make sense to use Mutation Testing?

Concurrency

1. Explain two conditions under which a deadlock can happen.
2. What is “non-deadlock” problem?
3. How would you test a concurrent program?
4. Is it a good idea to enforcing a schedule by using sleep() statements? What could go wrong?

Defect Models

1. What is the difference between failure model and fault model?
2. According to the lecture, when is a program considered “correct” or “incorrect”?
3. What is the failure domain of a program?
4. For each of the following explain if it is a fault or failure model: Division by zero, Combinatorial testing, Off-by-one, MBT, Sneak path, Operational profile.
5. What makes a fault model “good”?
6. Is there a fault model behind requirements-based testing? If not, why is it even used then?

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